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Aryendra et al.

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(54) **OVERLAY NETWORK FOR REAL-TIME PAYMENT NETWORKS**

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(57) **ABSTRACT**

Disclosed are various embodiments for facilitating payments between members of separate payment networks. A first instance of a supernetwork can receive a payment request from a source network hub connected to the first supernetwork instance and linked to a first payment network, the first payment request specifying an identifier for a recipient institution and an amount of the payment request. The first instance of the supernetwork can then query a participant status cache to identify a destination network hub linked to a second payment network associated with the recipient institution. Next, the first instance of the supernetwork can query a participant registry to identify a second

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Diagram illustrating a payment network system 100. The system includes a central Network Hub 106, which is connected to three Participant Systems: 103a, 103b, and 103c. The Network Hub 106 contains a Network ID 109, Participant Account(s) 113, Participant ID 116, and Balance 119. Participant System 103d is also connected to the Network Hub 106.

supernetwork instance connected to the destination network hub. Finally, the first instance of the supernetwork can forward the payment request to a second global transaction router hosted by a second supernetwork instance connected to the destination network hub.

20 Claims, 6 Drawing Sheets

(58) Field of Classification Search

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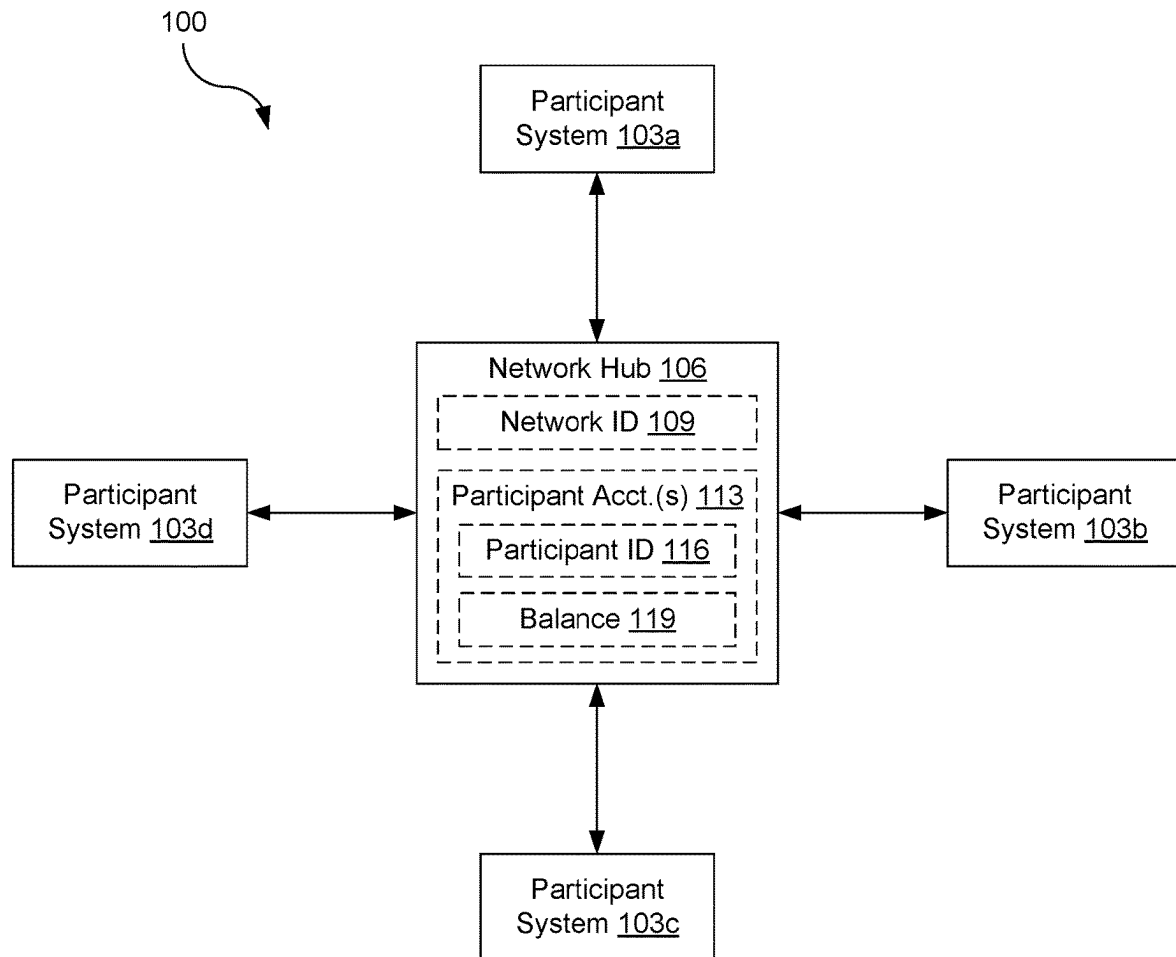
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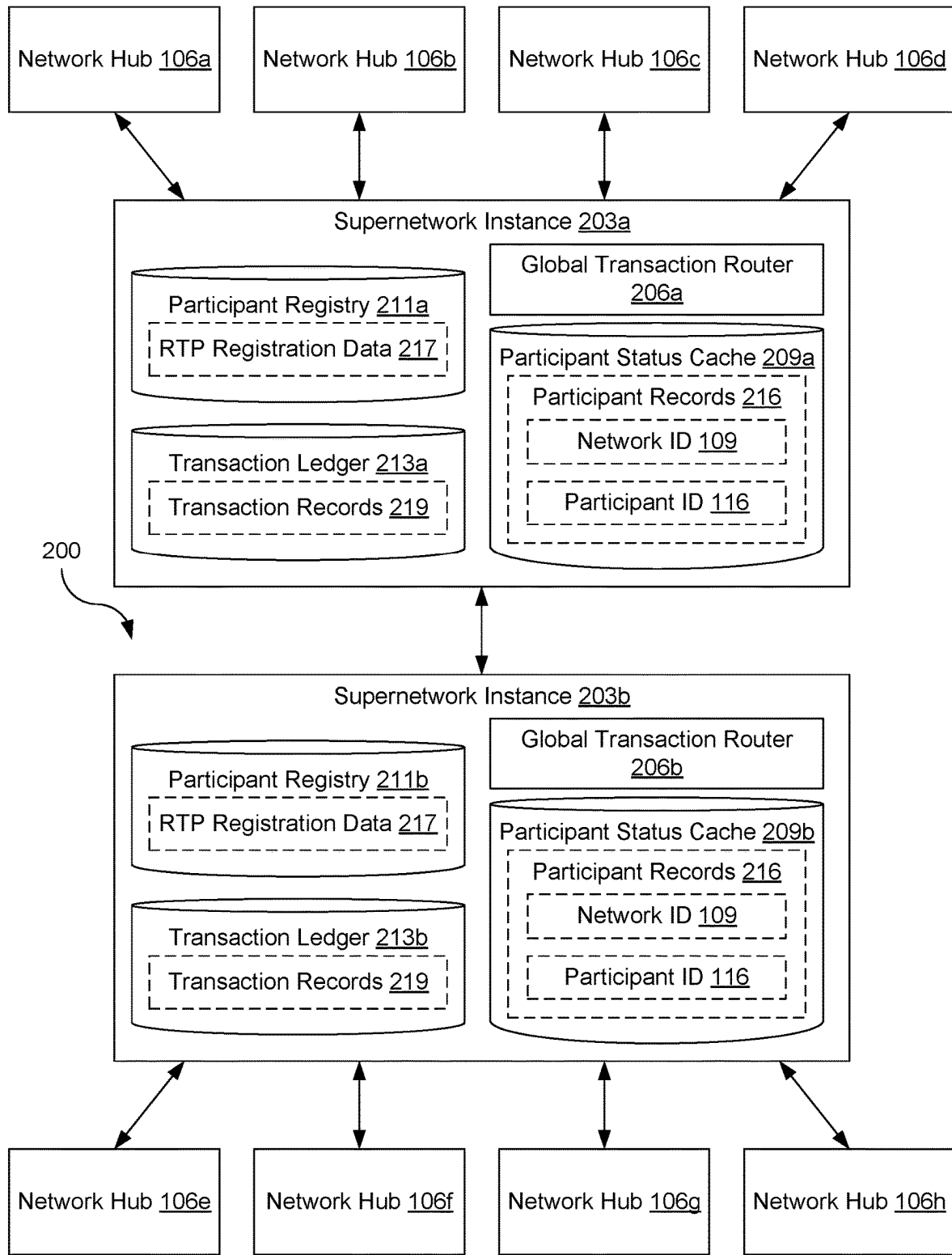
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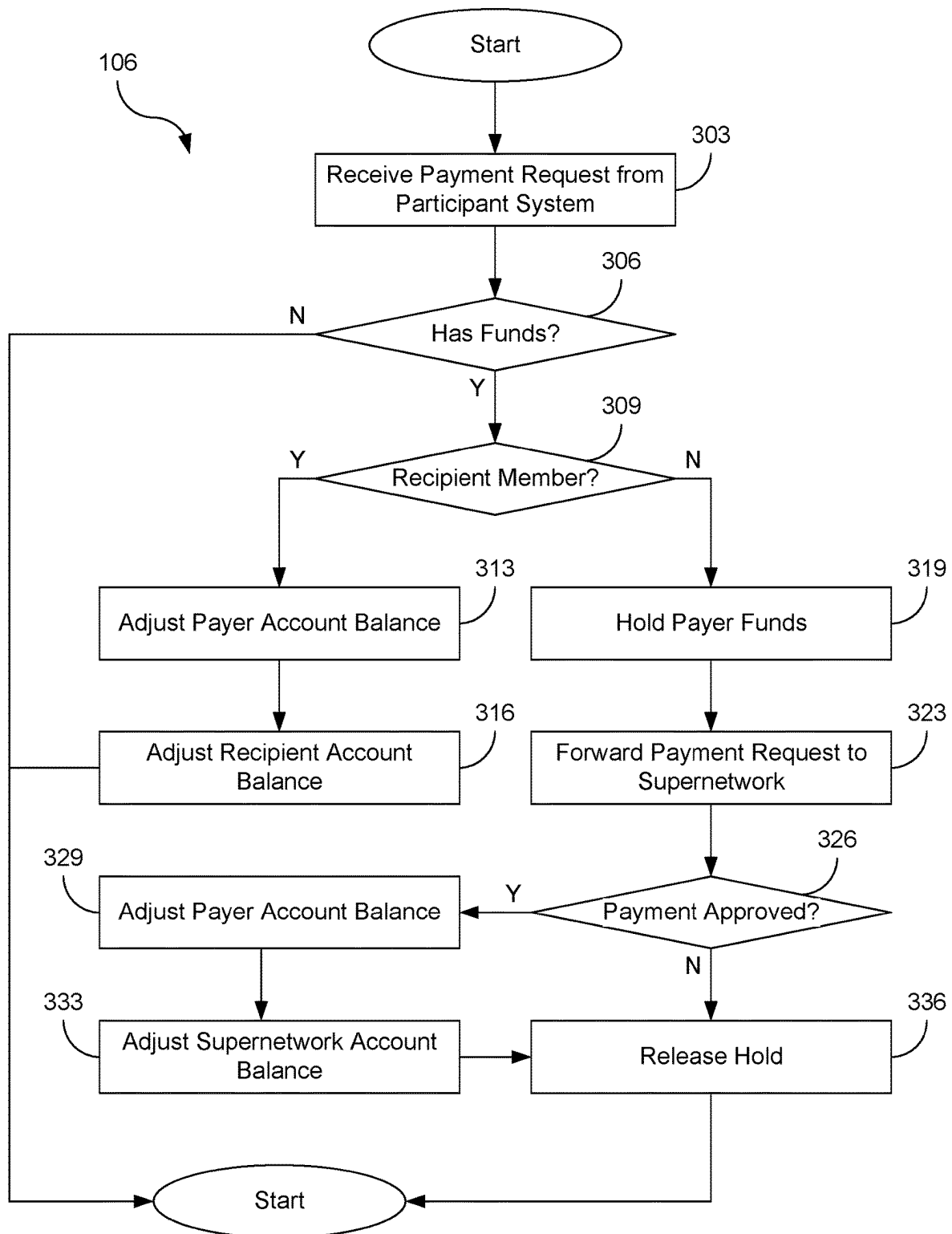
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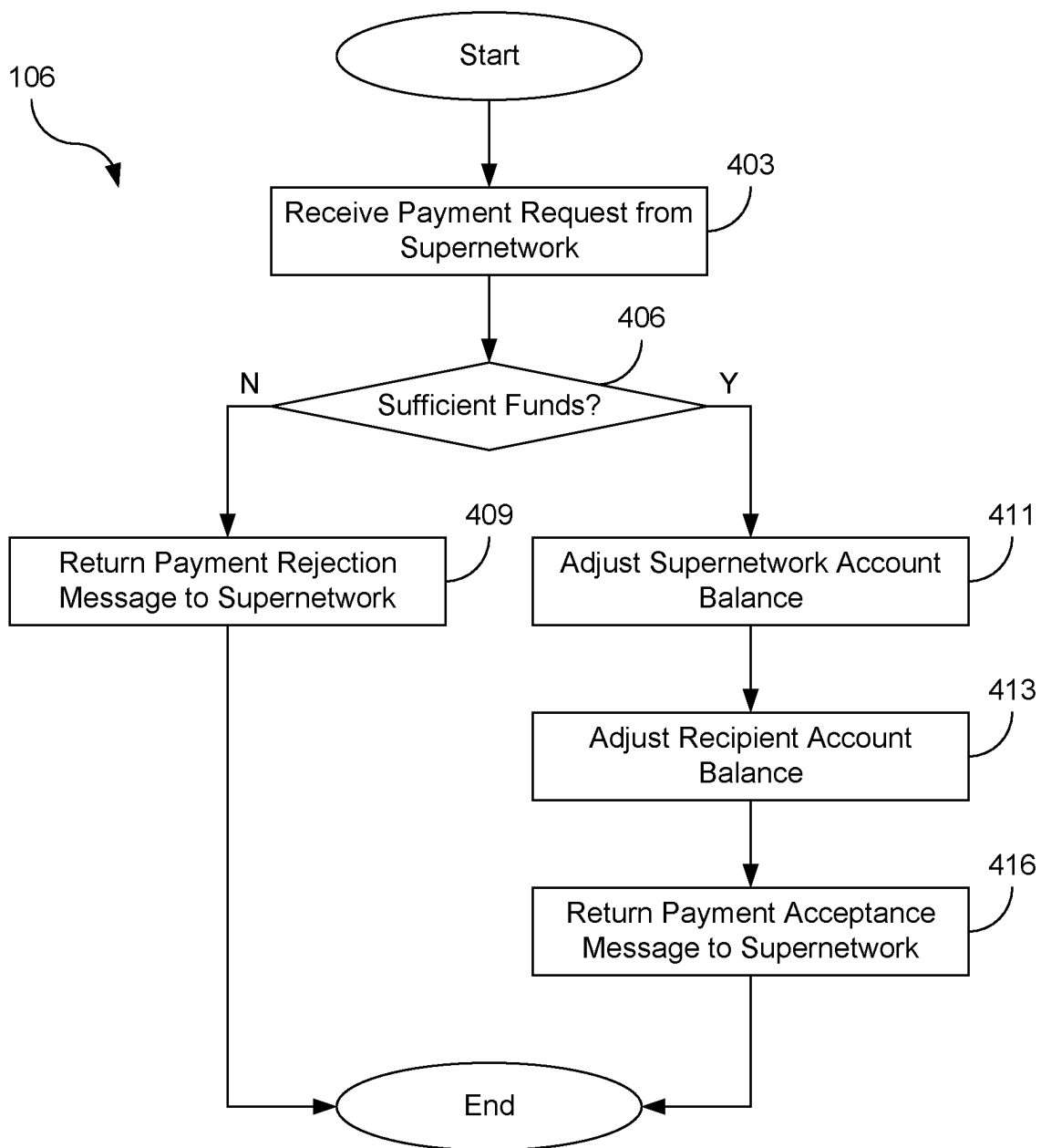
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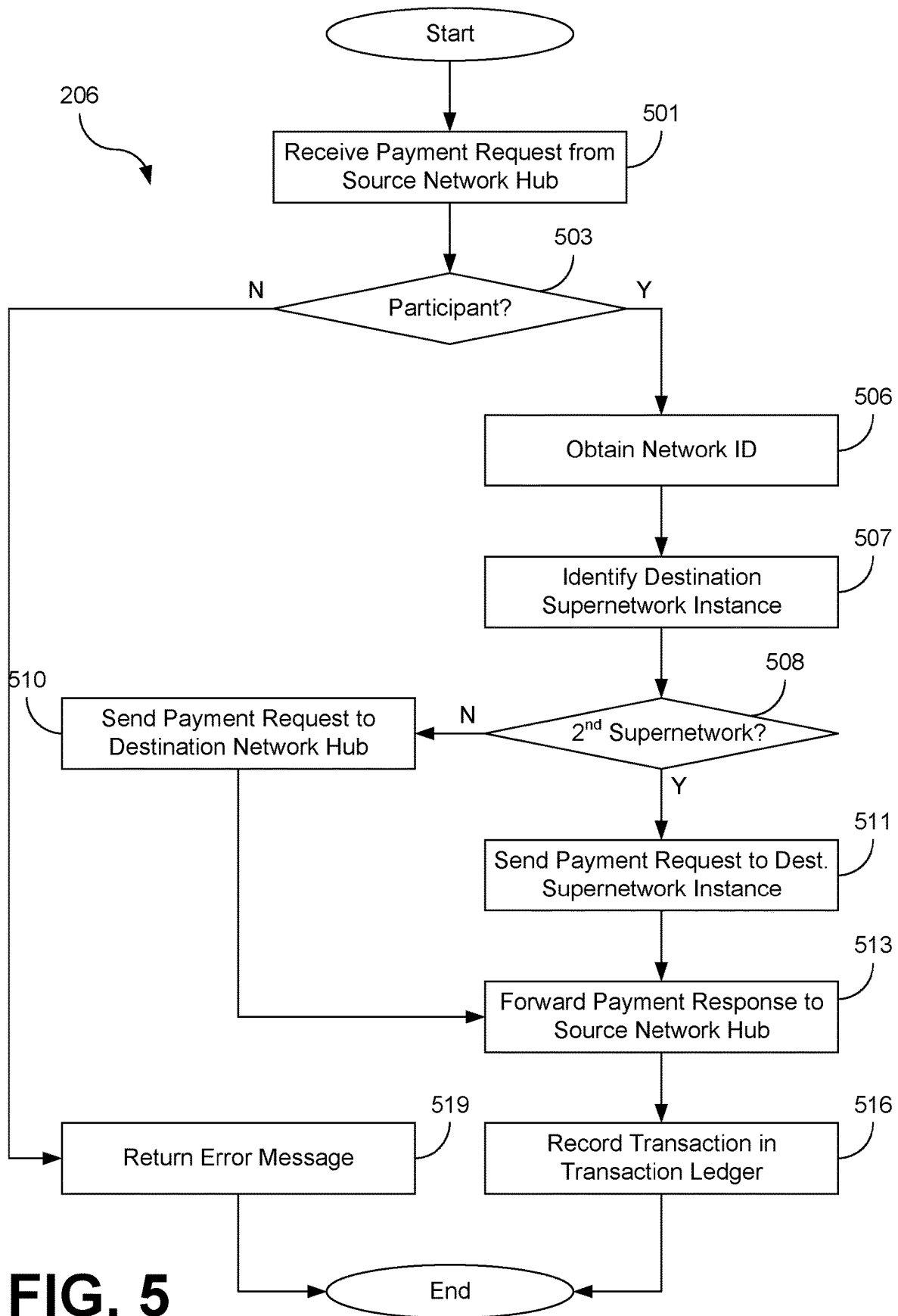
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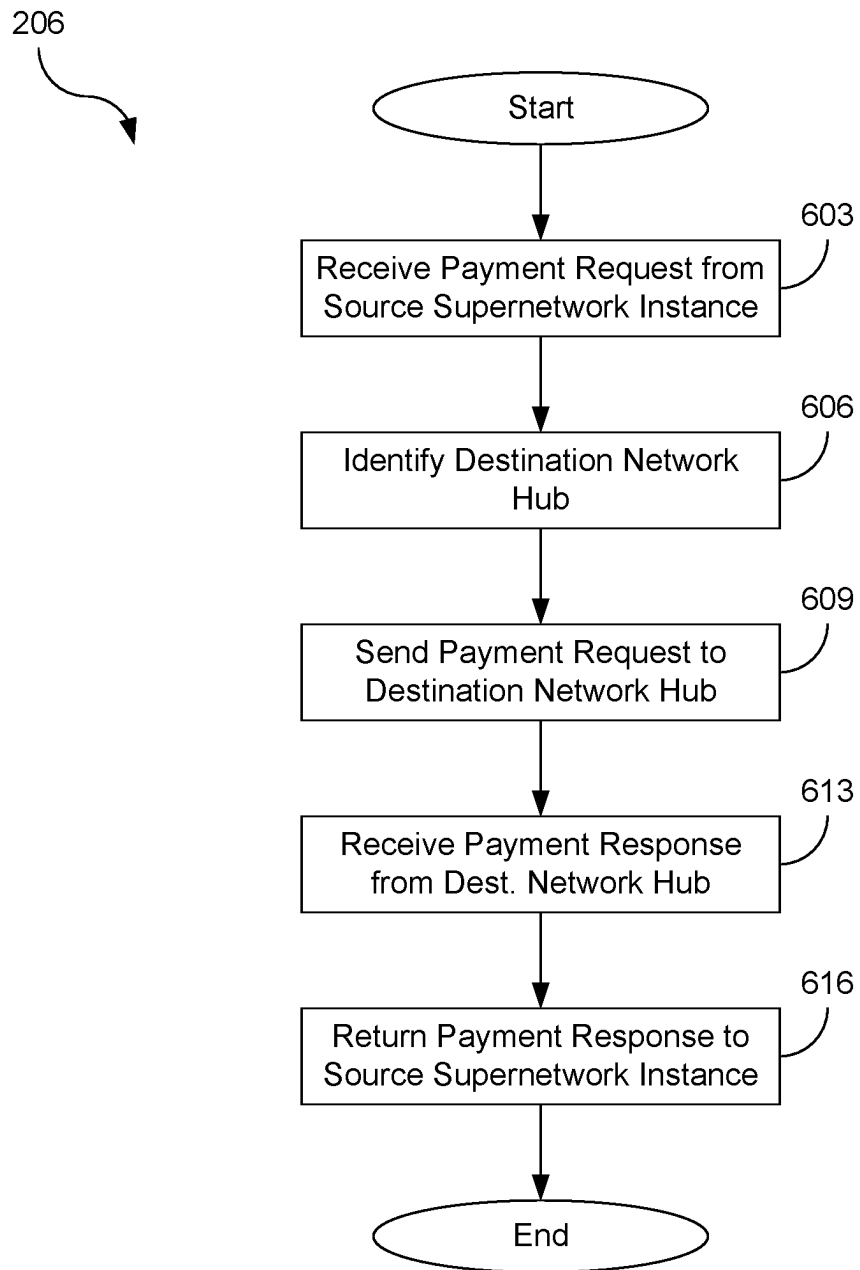
**FIG. 1**

**FIG. 2**



**FIG. 4**

**FIG. 5**

**FIG. 6**

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OVERLAY NETWORK FOR REAL-TIME PAYMENT NETWORKS

BACKGROUND

Financial institutions use real-time payment networks to send funds to other participating financial institutions in real-time or near real-time. Financial institutions may also be members of other types of payment networks. Moreover, financial institutions may be members of multiple payment networks to offer multiple payment options to customers and to increase the ability of the financial institution to make electronic payments to other financial institutions.

BRIEF DESCRIPTION OF THE DRAWINGS

Many aspects of the present disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale, with emphasis instead being placed upon clearly illustrating the principles of the disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

FIG. 1 is a drawing depicting a real time payment network according to various embodiments of the present disclosure.

FIG. 2 is a drawing of a supernetwork according to various embodiments of the present disclosure.

FIG. 3 is a flowchart illustrating one example of functionality implemented as portions of an application executed in a computing environment in the super network of FIG. 2 according to various embodiments of the present disclosure.

FIG. 4 is a flowchart illustrating one example of functionality implemented as portions of an application executed in a computing environment in the super network of FIG. 2 according to various embodiments of the present disclosure.

FIG. 5 is a flowchart illustrating one example of functionality implemented as portions of an application executed in a computing environment in the super network of FIG. 2 according to various embodiments of the present disclosure.

FIG. 6 is a flowchart illustrating one example of functionality implemented as portions of an application executed in a computing environment in the super network of FIG. 2 according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

Disclosed are various approaches for connecting and facilitating payments between different real-time payment networks. Financial institutions are often members of real-time payment (RTP) networks in order to allow real-time payments between the financial institutions. Generally, as long as two financial institutions are members of the same RTP network, they can make real-time payments with each other.

However, there are multiple RTP networks currently available. If two financial institutions are not members of the same RTP network, then they cannot make a real-time payment with each other. This can happen when multiple RTP networks are available within the same jurisdiction (e.g., FedNow and THE CLEARING HOUSE in the United States) or when financial institutions are located in different jurisdictions that provide different RTP networks due to regulatory differences and oversight. Accordingly, the various embodiments of the present disclosure solve the problem that occurs when a first financial institution that is a member of a real-time payment network wants to make a

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real-time payment to a second financial institution that is not a member of the real-time payment network.

In the following discussion, a general description of the system and its components is provided, followed by a discussion of the operation of the same. Although the following discussion provides illustrative examples of the operation of various components of the present disclosure, the use of the following illustrative examples does not exclude other implementations that are consistent with the principals disclosed by the following illustrative examples.

FIG. 1 is a schematic block diagram depicting an example of a real-time payment (RTP) network 100. An RTP network 100 is a payment rail or payment network that allows member institutions to make payments to each other with immediate availability at any time, in contrast to other payment rails or networks where payments may take several days to process and/or may only be made or processed on specific days or hours (e.g., only on business days and/or only during business hours). Examples of RTP networks 100 in the United States are THE CLEARING HOUSE RTP network offered by The Clearing House and the FEDNOW RTP network offered by the Federal Reserve. Other RTP networks 100 may be available in other jurisdictions.

The RTP 100 can have a number of components, such as one or more participant systems 103 (e.g., participant system 103a, participant system 103b, participant system 103c, participant system 103d, etc.) and a network hub 106. Participant systems 103 represent systems owned or operated by members of the RTP network 100, such as banks or other financial institutions that use the RTP network 100 to send and receive payments in real time. The network hub 106 can represent one or more computing systems and software services that receive and reconcile payment requests from participant systems 103.

Accordingly, the network hub 106 can store various data to allow it to facilitate payments from one network participant 103 to another network participant 103, as well as route payments from a network participant 103 of the RTP network 100 to another network participant 103 of another RTP network 100 using a supernetwork 200 (FIG. 2). This data could include a network identifier 109 and one or more participant accounts 113.

The network identifier 109 can represent any identifier that uniquely identifies an RTP network 100 with respect to another RTP network 100. The network identifier 109 can be used by the supernetwork 200 to identify or distinguish between individual RTP networks 100 when routing payments between RTP networks 100.

Participant accounts 113 can be used by the network hub 106 to track the amount of funds each participant system 103 has on deposit with the RTP network 100. Accordingly, each participant system 103 of a participant can be associated with a participant account 113. A participant account 113 can include information such as a participant identifier 116 and a balance 119. The participant identifier 116 can be any identifier that uniquely identifies a participant system 103, and therefore a participant, with respect to another participant system 103, and therefore another participant. The balance 119 can represent the amount of funds available in the participant account 113 to a respective participant. When a payment request is sent by a first participant system 103, the amount of the balance 119 in the participant account 113 associated with the first participant system 103 is reduced by the amount specified in the payment request. Meanwhile, the amount of the balance 119 in the participant account 113 of the second, recipient participant system 103 is increased by the amount specified in the payment request.

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Moreover, an operator of the supernetwork 200 can also maintain a participant account 113 with the RTP network 100. The participant account 113 of the supernetwork 200 can be used by the supernetwork 200 to facilitate payments between members of the RTP network 100 and members of other RTP networks 100, as described in further detail later.

FIG. 2 is a schematic block diagram depicting an example of a supernetwork 200 according to various embodiments of the present disclosure. The supernetwork 200 can be implemented to route payments between different RTP networks 100 (FIG. 2), thereby allowing a participant of a first RTP network 100 to make a real time payment to a participant of a separate, second RTP network 100. Accordingly, the supernetwork 200 can include one or more supernetwork instances 203, such as supernetwork instance 203a and supernetwork instance 203b, that form a peer-to-peer network with each other. Each supernetwork instance 203 can be in data communication with one or more network hubs 106, such as network hubs 106a, 106b, 106c, 106, 106e, 106f, 106g, and 106h.

Each supernetwork instance 203 can include a number of components. For example, a super network instance 203 can include a global transaction router 206, a participant status cache 209, a participant registry 211, and a transaction ledger 213. The global transaction router 206 can be executed to route payment requests between network hubs 106 of different RTP networks 100 connected to a supernetwork instance 203.

The participant status cache 209, participant registry 211, and the transaction ledger 213 are all representative of data stores and associated with the operation of the various applications or functional entities of the various embodiments of the present disclosure. The participant status cache 209, participant registry 211, and the transaction ledger 213 can be implemented as relational databases or non-relational databases such as object-oriented databases, hierarchical databases, hash tables or similar key-value data stores, as well as other data storage applications or data structures. Moreover, combinations of these databases, data storage applications, and/or data structures may be used together to provide a single, logical, data store. In many instances of the present disclosure, participant status cache 209, participant registry 211, and the transaction ledger 213 can be implemented as distributed, eventually consistent data stores in order to synchronize the data across multiple supernetwork instances 203. The participant status cache 209 can include one or more participant records 216. The participant registry 211 can store RTP registration data 217. Meanwhile the transaction ledger 213 can include one or more transaction records 219. Other data can also be stored in the participant status cache 209, participant registry 211, or the transaction ledger 213 as desired by various embodiments of the present disclosure. Moreover, while depicted separately, the data stored in the participant status cache 209, participant registry 211, and the transaction ledger 213 can be combined into one or more data stores in some implementations.

A participant record 216 represents a record of a participant system 103 that is a member of an RTP network 100. Each participant record 216 can include the network identifier 109 of the RTP network 100 that the participant system 103 is a member of, as well as the participant identifier 116 of the participant system 103 in the RTP network 100. If a participant or participant system 103 is a member of or participant in multiple RTP networks 100, then the participant or participant system 103 could be associated with multiple participant records 216. Other information can also

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be included in a participant record 216 as desired for particular implementations of the present disclosure.

RTP registration data 217 includes information about individual RTP networks 100 with a network hub 106 connected to a supernetwork instance 203 of the supernetwork. RTP registration data 217 can include a list of network hubs 106 or RTP networks 100 and the individual supernetwork instances 203 that the network hubs 106 are connect to or in data communication with. For example, RTP registration data 217 could map a network identifier 109 for an RTP network 100 to a particular supernetwork instance 203 (e.g., by using an instance identifier of the supernetwork instance 203). Other information regarding individual RTP networks 100 could also be stored in the RTP registration data 217 as desired for individual implementations of the present disclosure.

A transaction record 219 can represent a record of a transaction made between participants of two different RTP networks 100 within the supernetwork 200. Information stored in the transaction record 219 can include the participant identifier 116 and network identifier 109 of the payer, the participant identifier 116 and network identifier 109 of the payee, the amount of the transaction, as well as any other information that may be relevant to a particular embodiment of the present disclosure.

Next, a general description of the operation of the various components of the supernetwork 200 is provided. Although the following description provides merely an example of the operation of the supernetwork 200, and the interactions between individual components, other interactions and operations can also be performed by the various embodiments of the present disclosure. More detailed description of the operation of individual components is illustrated in the flowcharts of FIGS. 3-6.

To begin, a network hub 106 of an RTP network 100 can be configured to connect to a supernetwork instance 203 of the supernetwork 200. As part of the connection process, the network hub 106 can be configured to send and receive message to the supernetwork instance 203 using a supernetwork compliant message protocol. The network hub 106 could also be configured to translate payment messages from the format of the RTP network 100 serviced by the network hub 106 to the supernetwork compliant message protocol, and vice versa. Moreover, during the registration or first connection of the network hub 106 of the RTP network 100 with the supernetwork instance 203, the RTP registration data 217 for the RTP network 100 could be saved to the participant registry 211. For example, the network identifier 109 for the RTP network 100 could be saved in association with an instance identifier of the supernetwork instance 203 that the network hub 106 is connected to. The participant registry 211 could then replicate, distribute, or synchronize the RTP registration data with other participant registries 211 of other supernetwork instances 203.

Subsequently, the network hub 106 could provide a list of all participants in the RTP network 100 of the network hub 106 to the global transaction router 206 of the supernetwork instance 203. This could include the network identifier 109 of the RTP network 100 of the network hub 106, as well as the participant identifiers 116 of the participant systems 103 of the RTP network 100 of the network hub 106. In response, the global transaction router 206 could create and save a participant record 216 for each of the participants of the RTP network 100 of the network hub 106 to the participant status cache 209. The participant status cache 209 could then replicate, distribute, or synchronize the newly created par-

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participant records **216** to other participant status caches **209** of other supernetwork instances **203**.

Later, a network hub **106** of a first RTP network **100** (e.g., network hub **106a**) could receive a payment request from a participant system **103** to send a payment to a second participant system **103** that is part of a second RTP network **100** that uses a second network hub **106** (e.g., network hub **106h**). The first network hub **106a** could determine that the recipient of the transaction is not a member of the first RTP network **100**. In response, the first network hub **106a** could create and send a payment request to the global transaction router **206a** executed by the supernetwork instance **203a** that the network hub **106a** is connected to.

The global transaction router **206a** could evaluate the payment request to determine where to route the payment request. For example, the global transaction router **206a** could query the participant status cache **209a** to determine whether there is a participant record **216** matching a participant identifier **116** for the recipient. If a participant record **216** exists, the global transaction router **206a** could retrieve the network identifier **109** to determine which network hub **106** to route the payment request to. If the network identifier **109** fails to match the network identifier **109** of a network hub **106** connected to the supernetwork instance **203a**, then the global transaction router **206a** could query the RTP registration data **217** in the participant registry **211a** to determine which supernetwork instance **203** (e.g., supernetwork instance **203b**) the payment request should be routed to. The global transaction router **206a** could then send the payment request to the appropriate global transaction router **206b**.

The global transaction router **206b** can receive the payment request and evaluate it to determine which network hub **106** to route the payment request to. For example, the global transaction router **206b** could compare the network identifier **109** specified in the payment request to the network identifiers **109** of the network hubs **106** connected to the supernetwork instance **203b**. If the network identifier **109** in the payment request matches the network identifier **106** of a connected network hub **106**, such as network hub **106h**, then the global transaction router **206b** could forward the payment request to the recipient network hub **106h**.

The recipient network hub **106h** could return a response message, which either accepts or rejects the payment request, to the global transaction router **206b**. The global transaction router **206b** could then relay the response message to the global transaction router **206a**, and the global transaction router **206a** could relay the response message to the source network hub **106a**.

Referring next to FIG. 3, shown is a flowchart that provides one example of the operation of a portion of a network hub **106**. The flowchart of FIG. 3 provides merely an example of the many different types of functional arrangements that can be employed to implement the operation of the depicted portion of the network hub **106**. As an alternative, the flowchart of FIG. 3 can be viewed as depicting an example of elements of a method implemented within the RTP network **100** or the supernetwork **200**.

Beginning with block **303**, the network hub **106** can receive a payment request from a participant system **103**. The payment request can include information such as the participant identifier **116** of the recipient, the participant identifier **116** of the payee, the amount of the payment, and potentially other information.

Then, at block **306**, the network hub **106** can determine whether the balance **119** of the participant account **113** associated with the participant identifier **116** of the payee

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that submitted the payment request at block **303** has sufficient funds to settle the payment. If the balance **119** of the participant account **113** has insufficient funds to settle the payment, then the process can end. Optionally, the network hub **106** could send a rejection or error message to the participant system that submitted the payment request. However, if the balance **119** of the participant account **113** has sufficient funds, then the process can proceed to block **309**.

Moving on to block **309**, the network hub **106** can determine if the recipient identified in the payment request is a participant in the RTP network **100**. For example, the network hub **106** could search for a participant account **113** with a participant identifier **116** matching the participant identifier **116** specified in the payment request. If a matching participant account **113** is found, then the network hub **106** can determine that the recipient is a member of the RTP network **100** and the process can proceed to block **313**. However, if a matching participant account **113** is not found, then this would indicate that the recipient is not a member of the RTP network **100**. In this situation, the process could proceed to block **319**.

If the process proceeds to block **313**, the network hub **106** can adjust the account balance **119** of the participant account **113** of the payer. For example, the network hub **106** could deduct an amount of funds from the account balance **119** equal to the amount of funds specified in the payment request.

Next, at block **316**, the network hub **106**, can similarly adjust the account balance **119** of the participant account **113** of the recipient. For example, the network hub **106** could add an amount of funds to the account balance **119** of the recipient equal to the amount of funds specified in the payment request.

However, if the process instead proceeds to block **319**, the network hub **106** can place a hold on the account balance **119** of the participant account **113** of the payer that submitted the payment request at block **303**. This hold can be done to prevent double-spending of funds held in the participant account **113** of the payer while the network hub **106** forwards the payment request to a global transaction router **206** of a supernetwork instance **203**.

Then, at block **323**, the network hub **106** can forward the payment request to the global transaction router **206** of the supernetwork instance **203** that the network hub **106** is connected. In some implementations, the network hub **106** could create a new payment message or payment request that satisfies any protocol requirements of the supernetwork **200**. Generally, such a new payment message or payment request would include at least the same information that is included in the original payment, but be formatted in a standardized way that could be processed by the global transaction router **206**.

Moving on to block **326**, the network hub **106** can wait until it receives a payment response message from the global transaction router **206** of the supernetwork instance **203** that the network hub **106** is connected to. Once the payment response message is received, the network hub **106** can analyze the payment response message to determine if the payment request was accepted by the recipient network hub **106** or if the payment request was rejected. If the payment response message indicates that the payment request was accepted, then the process can proceed to block **329**. However, if the payment response message indicates that the payment request was rejected, then the process can skip to block **336**.

If the process proceeds to block **329**, the network hub **106** can adjust the payer account balance **119**. For example, the

network hub **106** could deduct an amount of funds from the account balance **119** equal to the amount specified in the payment request. In some instances, the payment response message could include additional transaction fees (e.g., transaction fees required by the supernetwork **200** or the recipient network hub **106** to process the payment). In these instances, the additional transaction fees could also be deducted from the account balance **119** of the participant account **113** of the payer.

Next, at block **333**, the network hub **106** can adjust the account balance **119** of a participant account **113** associated with the operator of the supernetwork **200**. For example, the network hub **106** could add an amount of funds equal to the amount specified in the payment request and any additional transaction fees to the account balance **119** of the participant account **113** of the supernetwork **200**.

Once the process proceeds to block **336**, the network hub **106** can release the hold on the account balance **119** of the participant account **113** of the payee. Then, the process could end.

Referring next to FIG. **4**, shown is a flowchart that provides one example of the operation of a portion of a network hub **106**. The flowchart of FIG. **4** provides merely an example of the many different types of functional arrangements that can be employed to implement the operation of the depicted portion of the network hub **106**. As an alternative, the flowchart of FIG. **4** can be viewed as depicting an example of elements of a method implemented within the RTP network **100** or the supernetwork **200**.

Beginning with block **403**, a network hub **106** can receive a payment request from a global transaction router **206** of a supernetwork instance **203** connected to the network hub **106**. For example, if a first network hub **106a** forwarded a payment request to the global transaction router **206a** using the process described in FIG. **3**, then a recipient network hub **106** (e.g., network hub **106b**) could receive a corresponding payment request from the global transaction router **206b** of the supernetwork instance **203b**.

Then, at block **406**, the network hub **106** can determine whether the account balance **119** of the participant account **113** associated with the operator of the supernetwork **200** has sufficient funds to complete the transaction. If there are insufficient funds (e.g., because the payment request is larger than the current account balance **119** of the supernetwork **200** within the RTP network **100**), then the process can proceed to block **409**. However, if there are sufficient funds to complete the payment request, then the process can proceed to block **411**.

If the process proceeds to block **409**, then the network hub **106** can generate a payment rejection message and return the payment rejection message to the global transaction router **206** of the supernetwork instance **203** that the network hub **106** is connected to. The payment rejection message can include the participant identifier **116** of the source of the payment and, potentially, the network identifier **109** of the source of the payment. In some instances, the payment rejection message could include a reason why the payment was rejected, while in other instances the reason for the rejection of the payment request could be omitted.

However, if the process proceeds to block **413**, then the network hub **106** can adjust the account balance **119** of a participant account **113** associated with the operator of the supernetwork **200**. For example, the network hub **106** could deduct an amount of funds equal to the amount specified in the payment request. The network hub **106** could also deduct any additional transaction fees from the account balance **119** of the participant account **113** of the operator of the super-

network **200** to compensate the RTP network **100** operator for the costs of processing the transaction.

Next, at block **413**, the network hub **106** can adjust the account balance **119** of the participant account **113** of the recipient. Accordingly, the network hub **106** could search for the participant account **113** matching the participant identifier **116** specified in the payment request. The network hub **106** could then add an amount of funds to the account balance **119** of the matching participant account **113** equal to the amount specified in the payment request.

Subsequently, at block **416**, the network hub **106** could generate and return a payment acceptance message to the global transaction router **206** of the supernetwork instance **203** that the network hub **106** is connected to. The payment acceptance message could include information such as a confirmation code or number, a confirmation of the amount deposited to the account balance **119** of the recipient, a timestamp indicating the time at which the recipient received the funds, the participant identifier **116** of the source of the payment and, potentially, the network identifier **109** of the source of the payment, as well as other information. The process can then subsequently end.

Referring next to FIG. **5**, shown is a flowchart that provides one example of the operation of a portion of the global transaction router **206**. The flowchart of FIG. **5** provides merely an example of the many different types of functional arrangements that can be employed to implement the operation of the depicted portion of the global transaction router **206**. As an alternative, the flowchart of FIG. **5** can be viewed as depicting an example of elements of a method implemented within the supernetwork **200**.

Beginning with block **501**, the global transaction router **206** can receive a payment request from a source network hub **106**. For example, the payment request could have been received as part of the process performed by the source network hub **106** at block **323**. The payment request can include information such as the network identifier **109** and participant identifier **116** of the participant making the payment, the participant identifier **116** of the recipient, the network identifier **109** of the participant (if known), the amount of the payment, and potentially other information depending on the particular implementation of the present disclosure.

Moving on to block **503**, the global transaction router **206** can determine whether the recipient of the payment request is a participant of an RTP network **100** with a network hub **106** connected to a supernetwork instance **203** of the supernetwork **200**. This network hub **106** would be the destination network hub **106** for the payment request. For example, the global transaction router **206** could search the participant status cache **209** to identify a participant record **216** with a matching participant identifier **116**. If a participant record **216** exists, then the process can proceed to block **506**. If no participant record **216** exists, then the process can instead skip to block **516**.

Then, at block **506**, the global transaction router **206** can obtain the network identifier **109** for the destination network hub **106** from the participant record **216** identified at block **503**.

Next, at block **507**, the global transaction router **206** can identify the supernetwork instance **203** which the destination network hub **106** associated with the destination network identifier **109** obtained at block **506** is connected to. For example, the global transaction router **206** could search the RTP registration data **217** in the participant registry **211** to identify the supernetwork instance **203** that is associated with the network identifier **109**. However, in some imple-

mentations, the global transaction router **206** could cache a list of network hubs **106** that it is connected to, in which case the global transaction router **206** could query its cache instead of the participant cache registry **211**.

Proceeding to block **508**, the global transaction router **206** can determine whether the destination network hub **106** for the payment request is connected to the supernetwork instance **203** (e.g., supernetwork instance **203a**) hosting the global transaction router **206**, or is connected to a second supernetwork instance **203** (e.g., supernetwork instance **203b**). This can be done by determining whether supernetwork instance **203** identified at block **507** is the same supernetwork instance **203** hosting the global transaction router **206**. If the destination network hub **106** is connected to the same supernetwork instance **203** that is hosting the global transaction router **206**, then the process can proceed to block **510**. However, if the destination network hub **106** is connected to a second supernetwork instance **203**, then the process can proceed to block **511**.

If the process proceeds to block **510**, the global transaction router **206** can send the payment request to the destination network hub **106** that is connected to the supernetwork instance **203** hosting the global transaction router **206**. The global transaction router **206** can then wait to receive a response from the destination network hub **106** regarding the payment status.

However, if the process proceeds to block **511**, the global transaction router **206** can send the payment request to the global transaction router **206** hosted by the supernetwork instance **203** identified at block **509**. The global transaction router **206** can then wait to receive a response from the second global transaction router **206** regarding the payment status.

Later, at block **513**, the global transaction router **206** can receive a payment response indicating the status of the payment request and return or forward it to the source network hub **106**. For example, the global transaction router **206** could search the participant status cache **209** to identify a participant record **216** with a matching participant identifier **116**. The global transaction router **206** could then determine the network identifier **109** for the destination of the message and determine that the network identifier **109** is for a network hub **106** (e.g., the source network hub **106**) connected to the supernetwork instance **203** hosting the global transaction router **206**. For example, the global transaction router **206** could query the RTP registration data in the participant cache registry **211** to determine that the destination network hub **106** is connected to the global transaction router **206**. However, in some implementations, the global transaction router **206** could cache a list of network hubs **106** that it is connected to, in which case the global transaction router **206** could query its cache instead of the participant cache registry **211**.

The global transaction router **206** can also cache or temporarily store the payment response for use at block **516**.

Then, at block **516**, the global transaction router **206** store or record the transaction in the transaction ledger **213**. For example, if the payment response received at block **513** were a payment acceptance message, then the global transaction router **206** could record a transaction record **219** in the transaction ledger **213** containing information such as a confirmation code or number, a confirmation of the amount deposited to the account balance **119** of the recipient, a timestamp indicating the time at which the recipient received the funds, the participant identifier **116** of the source of the payment and, potentially, the network identifier **109** of the source of the payment, as well as other informa-

tion. Likewise, if the payment response were a payment rejection message, then the global transaction router **206** could record a transaction record **219** in the transaction ledger **213** containing the participant identifier **116** of the source of the payment and, potentially, the network identifier **109** of the source of the payment. In some instances, the payment rejection message could include a reason why the payment was rejected, in which case the reason for the rejection could also be included in the transaction record **219**. The process could then end.

If the process proceeds to block **519**, the global transaction router **206** can generate and return an error message to the source network hub **106** indicating that the payment request could not be completed. In some implementations, the error message could include an indication of the problem (e.g., destination is not a member of a supported RTP network **100**, the supernetwork has insufficient funds at the destination, etc.). Once the error message is sent, the process can end.

Referring next to FIG. **6**, shown is a flowchart that provides one example of the operation of a portion of the global transaction router **206**. The flowchart of FIG. **6** provides merely an example of the many different types of functional arrangements that can be employed to implement the operation of the depicted portion of the global transaction router **206**. As an alternative, the flowchart of FIG. **6** can be viewed as depicting an example of elements of a method implemented within the supernetwork **200**.

Beginning with block **603**, the global transaction router **206** (e.g., global transaction router **206b**) of a second supernetwork instance (e.g., supernetwork instance **203b**) can receive a payment request from a first global transaction router **206** (e.g., global transaction router **206a**) hosted or executed by a first supernetwork instance (e.g., supernetwork instance **203a**). The payment request could be received as a result of the first global transaction router **206** determining that the second global transaction router **206** has a connection to the network hub **106** of the recipient of the payment. An example of this process has been previously discussed and illustrated by FIG. **5**.

Then, at block **606**, the global transaction router **206** identify the destination network hub **106**. For example, the global transaction router **206** could analyze the payment request to determine the participant identifier **116** of the recipient. The global transaction router **206** could then query the participant status cache to search for a participant record **216** with a matching participant identifier **116**. The global transaction router **206** could then retrieve the network identifier **109** from the participant record **216**. The global transaction router **206** could then query the RTP registration data in participant cache registry **211** to determine that the destination network hub **106** is connected to the global transaction router **206**. However, in some instances, the global transaction router **206** could cache a list of network hubs **106** that it is connected to, in which case the global transaction router **206** could query its cache instead of the participant cache registry **211**.

Next, at block **609**, the global transaction router **206** could then forward or otherwise send the payment request received at block **603** to the network hub **106** identified at block **606**.

Moving on to block **613**, the global transaction router **206** can receive a payment response from the destination network hub **106** that the payment request was forwarded to at block **609**. The payment response, as previously discussed, could be a payment acceptance, a payment rejection, or other payment status message.

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Subsequently, at block 616, the global transaction router 206 can return the payment response received at block 613 to the source supernetwork instance 203 from which the payment request was received at block 603. For example, the global transaction router 206 could search the participant status cache 209 to identify a participant record 216 with a matching participant identifier 116 specifying the source of the payment and, therefore, the destination for the payment response. The global transaction router 206 could obtain the network identifier 109 from the identified participant record 216. This could be skipped, however, in those instances where the payment response included the network identifier 109 identifying the destination of the payment response. The global transaction router 206 could then search the RTP registration data 217 in the participant registry 211 to identify the supernetwork instance 203 that is associated with the network identifier 109, and forward the payment response to the identified supernetwork instance 203. The process can subsequently end.

A number of software components previously discussed are stored in the memory of the respective computing devices and are executable by the processor of the respective computing devices. In this respect, the term “executable” means a program file that is in a form that can ultimately be run by the processor. Examples of executable programs can be a compiled program that can be translated into machine code in a format that can be loaded into a random access portion of the memory and run by the processor, source code that can be expressed in proper format such as object code that is capable of being loaded into a random access portion of the memory and executed by the processor, or source code that can be interpreted by another executable program to generate instructions in a random access portion of the memory to be executed by the processor. An executable program can be stored in any portion or component of the memory, including random access memory (RAM), read-only memory (ROM), hard drive, solid-state drive, Universal Serial Bus (USB) flash drive, memory card, optical disc such as compact disc (CD) or digital versatile disc (DVD), floppy disk, magnetic tape, or other memory components.

The memory includes both volatile and nonvolatile memory and data storage components. Volatile components are those that do not retain data values upon loss of power. Nonvolatile components are those that retain data upon a loss of power. Thus, the memory can include random access memory (RAM), read-only memory (ROM), hard disk drives, solid-state drives, USB flash drives, memory cards accessed via a memory card reader, floppy disks accessed via an associated floppy disk drive, optical discs accessed via an optical disc drive, magnetic tapes accessed via an appropriate tape drive, or other memory components, or a combination of any two or more of these memory components. In addition, the RAM can include static random access memory (SRAM), dynamic random access memory (DRAM), or magnetic random access memory (MRAM) and other such devices. The ROM can include a program-mable read-only memory (PROM), an erasable program-mable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), or other like memory device.

Although the applications and systems described herein can be embodied in software or code executed by general purpose hardware as discussed above, as an alternative the same can also be embodied in dedicated hardware or a combination of software/general purpose hardware and dedicated hardware. If embodied in dedicated hardware, each can be implemented as a circuit or state machine that

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employs any one of or a combination of a number of technologies. These technologies can include, but are not limited to, discrete logic circuits having logic gates for implementing various logic functions upon an application of one or more data signals, application specific integrated circuits (ASICs) having appropriate logic gates, field-programmable gate arrays (FPGAs), or other components, etc. Such technologies are generally well known by those skilled in the art and, consequently, are not described in detail herein.

The flowcharts show the functionality and operation of an implementation of portions of the various embodiments of the present disclosure. If embodied in software, each block can represent a module, segment, or portion of code that includes program instructions to implement the specified logical function(s). The program instructions can be embodied in the form of source code that includes human-readable statements written in a programming language or machine code that includes numerical instructions recognizable by a suitable execution system such as a processor in a computer system. The machine code can be converted from the source code through various processes. For example, the machine code can be generated from the source code with a compiler prior to execution of the corresponding application. As another example, the machine code can be generated from the source code concurrently with execution with an interpreter. Other approaches can also be used. If embodied in hardware, each block can represent a circuit or a number of interconnected circuits to implement the specified logical function or functions.

Although the flowcharts show a specific order of execution, it is understood that the order of execution can differ from that which is depicted. For example, the order of execution of two or more blocks can be scrambled relative to the order shown. Also, two or more blocks shown in succession can be executed concurrently or with partial concurrence. Further, in some embodiments, one or more of the blocks shown in the flowcharts can be skipped or omitted. In addition, any number of counters, state variables, warning semaphores, or messages might be added to the logical flow described herein, for purposes of enhanced utility, accounting, performance measurement, or providing troubleshooting aids, etc. It is understood that all such variations are within the scope of the present disclosure.

Also, any logic or application described herein that includes software or code can be embodied in any non-transitory computer-readable medium for use by or in connection with an instruction execution system such as a processor in a computer system or other system. In this sense, the logic can include statements including instructions and declarations that can be fetched from the computer-readable medium and executed by the instruction execution system. In the context of the present disclosure, a “computer-readable medium” can be any medium that can contain, store, or maintain the logic or application described herein for use by or in connection with the instruction execution system. Moreover, a collection of distributed computer-readable media located across a plurality of computing devices (e.g., storage area networks or distributed or clustered filesystems or databases) may also be collectively considered as a single non-transitory computer-readable medium.

The computer-readable medium can include any one of many physical media such as magnetic, optical, or semiconductor media. More specific examples of a suitable computer-readable medium would include, but are not limited to, magnetic tapes, magnetic floppy diskettes, magnetic hard

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drives, memory cards, solid-state drives, USB flash drives, or optical discs. Also, the computer-readable medium can be a random access memory (RAM) including static random access memory (SRAM) and dynamic random access memory (DRAM), or magnetic random access memory (MRAM). In addition, the computer-readable medium can be a read-only memory (ROM), a programmable read-only memory (PROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), or other type of memory device.

Further, any logic or application described herein can be implemented and structured in a variety of ways. For example, one or more applications described can be implemented as modules or components of a single application. Further, one or more applications described herein can be executed in shared or separate computing devices or a combination thereof. For example, a plurality of the applications described herein can execute in the same computing device, or in multiple computing devices in the same computing environment.

Disjunctive language such as the phrase “at least one of X, Y, or Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to present that an item, term, etc., can be either X, Y, or Z, or any combination thereof (e.g., X; Y; Z; X or Y; X or Z; Y or Z; X, Y, or Z; etc.). Thus, such disjunctive language is not generally intended to, and should not, imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

It should be emphasized that the above-described embodiments of the present disclosure are merely possible examples of implementations set forth for a clear understanding of the principles of the disclosure. Many variations and modifications can be made to the above-described embodiments without departing substantially from the spirit and principles of the disclosure. All such modifications and variations are intended to be included herein within the scope of this disclosure and protected by the following claims.

Therefore, the following is claimed:

1. A system, comprising:

a computing device comprising a processor and a memory; and

a first supernetwork instance of a supernetwork configured to allow a first participant of a first payment network to make a payment to a second participant of a second payment network, the first supernetwork instance comprising a first global transaction router, a first participant registry, a first transaction ledger, and a first participant status cache, wherein the first supernetwork instance is in data communication with one or more network hubs and at least a second supernetwork instance,

wherein the first global transaction router hosted by the first supernetwork instance comprises machine-readable instructions stored in the memory that, when executed by the processor, cause the computing device to at least:

receive a first payment request from a source network hub connected to the first supernetwork instance and linked to the first payment network, the first payment request specifying at least a participant identifier for a recipient institution and an amount of the first payment request;

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query the first participant status cache to identify a participant record associated with the participant identifier of the recipient institution;

obtain, from the participant record, a network identifier for a destination network hub linked to the second payment network associated with the recipient institution;

query the first participant registry to identify the second supernetwork instance connected to the destination network hub associated with the network identifier; forward the first payment request to a second global transaction router hosted by the second supernetwork instance connected to the destination network hub;

receive, from the second global transaction router or the destination network hub, a payment response indicating a status of the payment request; and record, in the first transaction ledger, a transaction record comprising at least the payment response.

2. The system of claim 1, wherein the first global transaction router further causes the computing device to at least: receive a second payment request from the second global transaction router;

query the first participant registry to determine that the second payment request is destined for a second destination network hub connected to the first supernetwork instance; and

forward the second payment request to the second destination network hub.

3. The system of claim 2, wherein the first global transaction router further causes the computing device to at least: receive the payment response from the destination network hub;

query the first participant registry to determine that the payment response is destined for a network hub associated with the second supernetwork instance hosting the second global transaction router; and

forward the payment response to the second global transaction router.

4. The system of claim 1, wherein the first global transaction router further causes the computing device to at least: receive the payment response from the second global transaction router; and

query the first participant status cache to identify the source network hub as the recipient of the payment response;

query the first participant registry or the first participant status cache to determine that the source network hub is connected to the first supernetwork instance; and forward the payment response to the source network hub.

5. The system of claim 4, wherein the first global transaction router further causes the computing device to at least: retrieve transaction information from the payment response; and

record the transaction record in the first transaction ledger, the transaction record comprising the transaction information.

6. The system of claim 4, wherein the payment response is a payment acceptance message.

7. The system of claim 4, wherein the payment response is a payment rejection message.

8. A method, comprising:

receiving, by a first global transaction router hosted by a first supernetwork instance, a first payment request from a source network hub connected to the first supernetwork instance and linked to a first payment network, the first payment request specifying at least a

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participant identifier for a recipient institution and an amount of the first payment request;

querying, by the first global transaction router hosted by the first supernetwork instance, a participant status cache to identify a participant record associated with the participant identifier of the recipient institution;

obtaining, by the first global transaction router hosted by the first supernetwork instance, a network identifier for a destination network hub linked to a second payment network associated with the recipient institution from the participant record;

querying, by the first global transaction router hosted by the first supernetwork instance, a participant registry to identify a second supernetwork instance connected to the destination network hub associated with the network identifier;

forwarding, by the first global transaction router hosted by the first supernetwork instance, the first payment request to a second global transaction router hosted by a second supernetwork instance connected to the destination network hub;

receiving, by the first global transaction router hosted by the first supernetwork instance, a payment response; and

recording, by the first global transaction router hosted by the first supernetwork instance, a transaction record comprising at least the payment response.

9. The method of claim 8, further comprising:

receiving, by the first global transaction router hosted by the first supernetwork instance, a second payment request from the second global transaction router;

querying, by the first global transaction router hosted by the first supernetwork instance, the participant registry to determine that the second payment request is destined for a second destination network hub connected to the first supernetwork instance; and

forwarding, by the first global transaction router hosted by the first supernetwork instance, the second payment request to the second destination network hub.

10. The method of claim 9, further comprising:

receiving, by the first global transaction router hosted by the first supernetwork instance, a payment response from the destination network hub;

querying, by the first global transaction router hosted by the first supernetwork instance, the participant registry to determine that the payment response is destined for a network hub associated with the second supernetwork instance hosting the second global transaction router; and

forwarding, by the first global transaction router hosted by the first supernetwork instance, the payment response to the second global transaction router.

11. The method of claim 8, further comprising:

receiving, by the first global transaction router hosted by the first supernetwork instance, a payment response from the second global transaction router; and

querying, by the first global transaction router hosted by the first supernetwork instance, a participant status cache to identify the source network hub as the recipient of the payment response;

querying, by the first global transaction router hosted by the first supernetwork instance, a participant registry or a cache to determine that the source network hub is connected to the first supernetwork instance; and

forward, by the first global transaction router hosted by the first supernetwork instance, the payment response to the source network hub.

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12. The method of claim 11, further comprising:

retrieving, by the first global transaction router hosted by the first supernetwork instance, transaction information from the payment response; and

recording, by the first global transaction router hosted by the first supernetwork instance, a transaction record in a transaction ledger, the transaction record comprising the transaction information.

13. The method of claim 11, wherein the payment response is a payment acceptance message.

14. The method of claim 11, wherein the payment response is a payment rejection message.

15. A non-transitory, computer-readable medium, comprising a first global transaction router hosted by a first supernetwork instance, the first global transaction router comprising machine-readable instructions that, when executed by a processor of a computing device, cause the computing device to at least:

receive a first payment request from a source network hub connected to the first supernetwork instance and linked to a first payment network, the first payment request specifying at least a participant identifier for a recipient institution and an amount of the first payment request;

query a participant status cache to identify a participant record associated with the participant identifier of the recipient institution;

obtain, from the participant record, a network identifier for a destination network hub linked to a second payment network associated with the recipient institution;

query a participant registry to identify a second supernetwork instance connected to the destination network hub associated with the network identifier;

forward the first payment request to a second global transaction router hosted by a second supernetwork instance connected to the destination network hub;

receive, from the second global transaction router or the destination network hub, a payment response indicating a status of the payment request; and

record, in a transaction ledger, a transaction record comprising at least the payment response.

16. The non-transitory, computer-readable medium of claim 15, wherein the first global transaction router further causes the computing device to at least:

receive a second payment request from the second global transaction router;

query the participant registry to determine that the second payment request is destined for a second destination network hub connected to the first supernetwork instance; and

forward the second payment request to the second destination network hub.

17. The non-transitory, computer-readable medium of claim 16, wherein the first global transaction router further causes the computing device to at least:

receive a payment response from the destination network hub;

query the participant registry to determine that the payment response is destined for a network hub associated with the second supernetwork instance hosting the second global transaction router; and

forward the payment response to the second global transaction router.

18. The non-transitory, computer-readable medium of claim 15, wherein the first global transaction router further causes the computing device to at least:

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receive a payment response from the second global transaction router; and

query a participant status cache to identify the source network hub as the recipient of the payment response;

query a participant registry or a cache to determine that the source network hub is connected to the first super-network instance; and

forward the payment response to the source network hub.

19. The non-transitory, computer-readable medium of claim **18**, wherein the first global transaction router further causes the computing device to at least:

retrieve transaction information from the payment response; and

record a transaction record in a transaction ledger, the transaction record comprising the transaction information.

20. The non-transitory, computer-readable medium of claim **18**, wherein the payment response is a payment rejection message or a payment acceptance message.

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